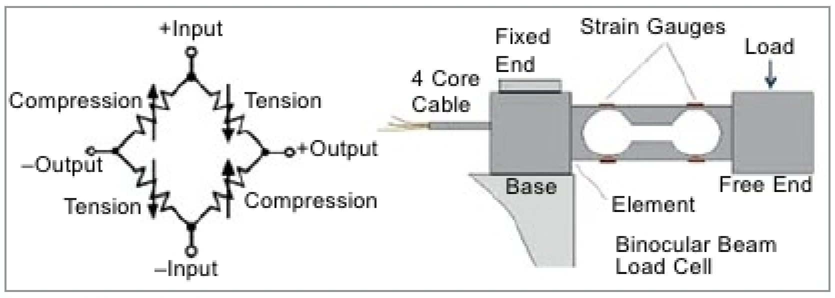

Fig. 4: Load cell

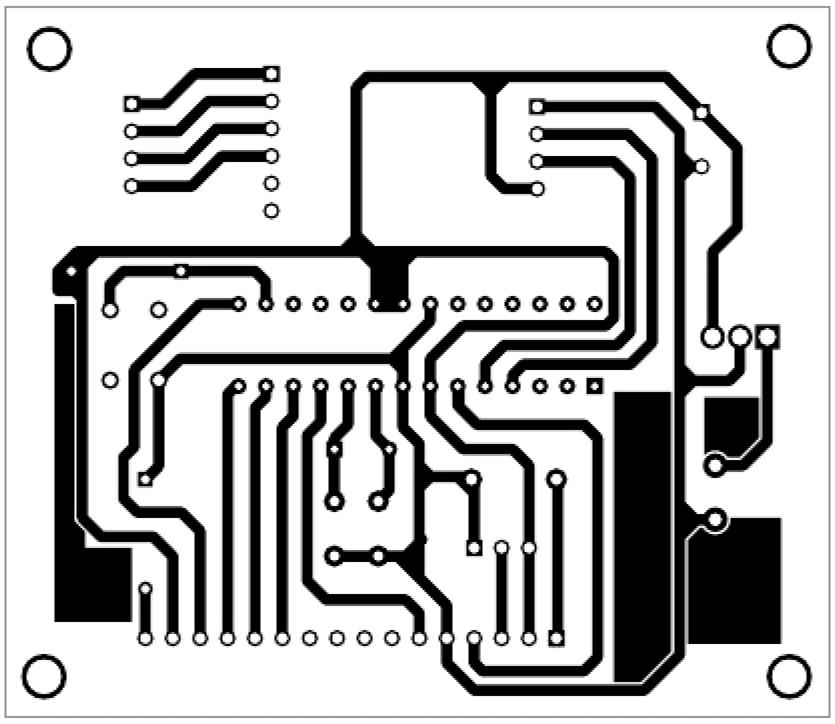
Fig. 5: Actual-size PCB layout of the weighing machine circuit

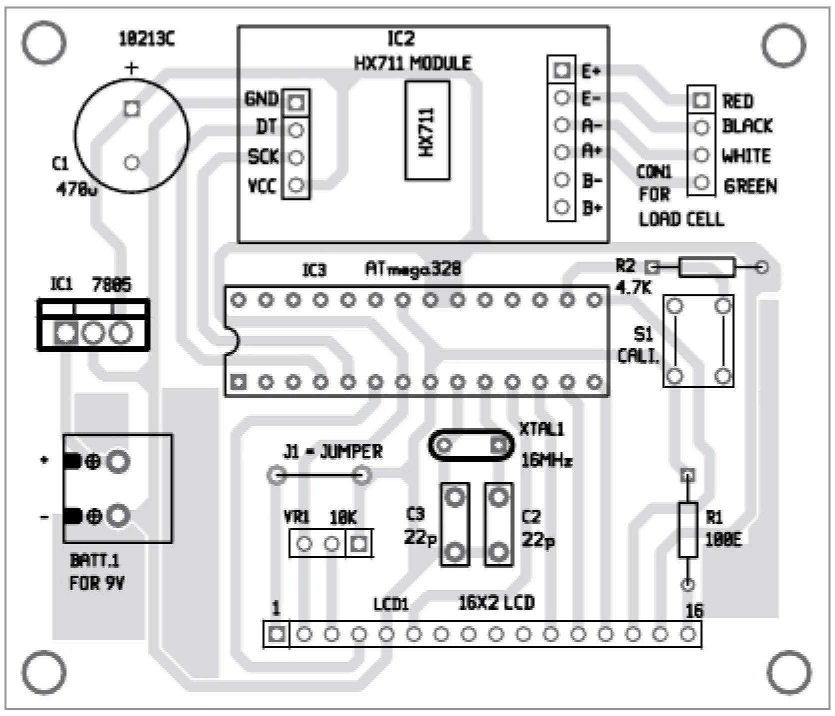

Fig. 6: Components side of the PCB

The first line takes 25 readings and averages it five times using the second line of code. To increase the speed of reading, reduce the number of averages. If input is blank, it will not take any average.

Software is built on Arduino onboard principle. After uploading the code into ATmega328 using Arduino Uno, take ATmega328 IC out and put into the PCB board.

Initial readings may have errors, as the system is not yet calibrated. Press push button S1 connected to pin 16 (PB2) of IC3 once and the system will self-calibrate. Initial reading of 0.0 will be shown on LCD1. The zero_set figure will then be stashed into the inbuilt EEPROM of Arduino.

Next time when you restart it after switch off operation, zero_set reading will be read from the EEPROM and internal calibration will be adjusted accordingly. It will be shown on the top line of LCD1. The second line will show 0.0.

If the readings reduce with increase in load, reverse the white and green wires, and the problem will be solved.

Construction and testing

An actual-size PCB layout for the weighing machine is shown in Fig. 5 and its components layout in Fig. 6. After assembling the circuit, enclose it in a suitable box. Attach one end of the load cell in a fixed frame. The other end should be used for placing the load or object for weight measurement.

Calibration. Calibration is an important part of measuring devices. For HX711 there is one calibration factor 7050 that works well for Adafruit modules. Therefore it has not been changed. The other is zero_set factor for which provision should be made to do it dynamically during operations. After each calibration, zero_set figure goes into EEPROM memory of Arduino. Next time when you restart the circuit, this value will be fetched from EEPROM and zero_set achieved.

However, to change zero-set reading, like for weighing milk (in a container), you would not be interested in measuring the container weight and then deducting it. So, put the container on the scale and press S1. Container weight will be added to zero_set reading and a new zero reading will be shown. Now, pour milk in the container and read the weight. To remove the container weight from zero_set reading, remove the container and press S1 again.

Multiplying factor. Change this factor suitably by putting a known weight on the load tray the first time. Once this is done, your machine will work fine. This is explained in the source code very clearly.

Fitting of the scale. If you look at the load cell, it has strain gauge attached on all three sides with suitable strain-relieving frames inbuilt in the design. Attach a suitable tray on the load end of the load cell for holding the object for weight measurement. Put a thick washer between the base plate of the tray and the load cell, and tighten firmly. Your weighing machine is ready for use. EFY

PARTS LIST

Semiconductors:	
IC1	- 7805, 5V voltage regulator
IC2	- HX711 module
IC3	- ATmega328 microcontroller with bootloader
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 100-ohm
R2	- 4.7-kilo-ohm
VR1	- 10-kilo-ohm potmeter
Capacitors:	
C1	- 470µF, 16V electrolytic
C2, C3	- 22pF ceramic disk
Miscellaneous:	
LCD1	- 16x2 LCD
S1	- Tactile switch
XTAL1	- 16MHz crystal
BATT.1	- 9V battery
	- Load cell up to 5kg
	- Arduino Uno board
	- 2-pin terminal connector
	- Nuts, bolts and washer

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*


Somnath Bera is an avid user of open source software. Professionally, he is a thermal power expert and works as additional general manager at NTPC Ltd